OIT 367 – BUSINESS INTELLIGENCE FROM BIG DATA

Final Project:

Subscriber Churn Analysis for iFit

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1. INTRODUCTION

1.1. Summary

We analyzed iFit subscriber churn to improve retention and profitability. Using customer data, we identified key churn predictors and built a forecasting model to enable proactive retention strategies that protect subscription revenue and maximize customer lifetime value.

1.2 Problem Statement

Our analysis answers: What drives subscriber churn, and how can we predict which customers have high cancellation risk?

While iFit has a large user base, **retention remains a key challenge** affecting revenue. This analysis serves management and customer success teams tasked with reducing attrition and maximizing revenue.

1.3. Findings and Recommendations

Our churn analysis revealed key drivers of churn:

1. **Equipment investment**: higher MSRP correlates with retention. Luxury equipment owners (>$2000) churn at half the rate (44.1%) of budget equipment owners (<$500, 83.1%).
2. **Demographics**: Younger subscribers (under 35) churn at higher rates (70-75%) than older subscribers (51-55%).
3. **High-risk personas**: "Uncommitted Experimenters" and "Quick Dropouts" show **79%+ churn rates** while "Fitness Diehards" maintain **~1.5% churn**.
4. **External trends**: Google data shows **seasonal subscription spikes** with many users disengaging within three months.

We identified concerning data patterns suggesting recording issues: all churn occurs in TRAIN tier (96.5%) with 0% in PREMIUM/PRO, and mostly in MULTI_YEAR plans (94.4%) with 0% in MONTHLY plans. These statistically improbable distributions contradict industry norms, indicating potential selection bias, data truncation, or inconsistent recording practices across segments.

Figure 1: User Tenure Distribution *This bar chart shows the distribution of subscribers across tenure periods. Nearly half (47.9%) of users remain subscribers for over a year (366+ days), while 20.2% churn between 31-90 days. Other significant tenure brackets include 91-180 days (13.4%), 181-365 days (11.6%), and 0-30 days (6.2%). This bimodal distribution suggests critical intervention periods during early onboarding and around the one-year mark.*

Our predictive model achieved 86.7% accuracy with 96.9% recall, effectively identifying at-risk subscribers. We recommend iFit develop retention programs targeting budget equipment owners and younger demographics, leverage predictive models to proactively identify high-risk customers, create persona-based retention offers, and collect data to further investigate plan and payment method impacts. The recommendations outlined in this report could **preserve $1.6 million/year.**

2. DATA

2.1. Overview

We leveraged three data sources: (1) dataset of iFit subscribers, (2) social media content from the r/iFit subreddit, and (3) Google Trends data with market patterns.

Subscriber Dataset: 1,049,810 subscribers with the following components:

User Information	User identifiers
	Demographic information (age, gender)
	Location (country)
Subscription Information	Subscription tier (TRAIN, PRO, PREMIUM)

	Payment method (MONTHLY, ANNUAL, MULTI_YEAR)
	Start date
	Churn date
	Full access flag
Equipment Information	Equipment type
	Console type
	Screen size
	MSRP

Reddit Dataset: we scraped 950 posts from the r/iFit subreddit, capturing discussions about products, workouts, and apps, for context about user experiences and churn motivations.

Google Trends Data for terms like "iFit," "home workouts," and "fitness app" to understand how external factors and seasonal patterns influence subscription behavior, and identify market-level patterns explaining the timing-based churn observed.

2.2. Defining "Success"

We defined churn by the presence of a "CHURN_DATE". The dataset shows a 61.15% churn rate (641,979 of 1,049,810 subscribers).

To prevent timing bias, we used a temporally-aware train-test split based on subscription start date instead of random sampling. This avoids misclassifying recently acquired customers as "retained" due to insufficient time to churn. We created prediction windows (30, 60, 90, 180, and 365 days), offering actionable predictions while maintaining sufficient training data.

2.3 Pre-Processing / Feature Engineering

We conducted data cleaning, removing invalid records where necessary, and created engineered features:

Temporal Variables (for exploratory data analysis)	Tenure (days between start and churn/current date)
	Tenure buckets (0-30 days, 31-90 days, etc.)
Categorization variables	Equipment price tiers (Budget, Mid-range, Premium, Luxury)
	Age groups (Under 25, 25-34, 35-44, 45-54, 55+)

	Equipment categories

2.4. Data Limitations

Several limitations affected our analysis:

1. **Selection bias**: dataset only includes actual subscribers, excluding those who considered but didn't purchase.
2. Some metrics measured at current date rather than subscription time, creating potential **temporal bias.**
3. **Limited revenue information** for precise LTV calculations.
4. **Limited behavioral data**: no usage patterns included, which likely influences retention.
5. **Limited availability of churn data** for different subscription tiers and payment methods.

3. METHODS

We combined exploratory data analysis, predictive modeling, survival analysis, and customer segmentation to understand churn dynamics and develop actionable insights.

3.1. Exploratory Analysis

We explored churn patterns across subscription tiers, payment methods, demographics, and equipment types, revealing significant variations between customer segments and highlighting key risk factors.

3.2. Predictive Modeling

We developed a Random Forest classifier as our primary model, with:

- Feature preprocessing: standardized numerical features, one-hot encoded categorical variables
- Class imbalance handling (SMOTE)
- Temporal validation: temporally-aware train-test split, not random sampling
- Threshold optimization to balance precision and recall: classifier validated on a test set with most recent 20% data, ensuring predictions would work on future data
- Feature importance analysis identifying predictive factors

3.3 Survival Analysis

We applied Kaplan-Meier estimators to understand customer lifecycle and critical high-churn periods, estimating survival probabilities over time across customer segments, comparing retention curves across factors, and identifying how long customers remain before churning.

3.4. Customer Segmentation

We implemented K-means clustering to identify customer personas based on demographics, behavior, and equipment. We selected k=6 as it provided the most interpretable and actionable segments. This allowed us to identify customer groups with varying churn risks and lifetime values.

3.5. Reddit Sentiment Analysis

We leveraged NLP to analyze Reddit data, employing VADER and TextBlob NLP with keyword-based filtering, examining 950 posts from the r/iFit subreddit. Initially, we used NLP to classify sentiment but found limitations in detecting nuanced dissatisfaction. We then focused on "cancel" mentions and manually categorized 22 relevant posts into five complaint types. Our metrics included sentiment distribution (sometimes misleadingly neutral), cancellation categories, and post engagement (comments and interactions).

3.6. Google Trends Analysis

We examined Google Trends data to determine how trends like New Year's fitness resolutions impact behavior. We collected 5-year historical search data for "iFit," "home workouts," and "fitness app" terms at weekly intervals to capture seasonal patterns. We compared search volume to iFit's data, measured correlation between search volume and sign-ups, and identified seasonal patterns in search interest and subscriptions.

4. RESULTS

4.1. Overall Churn Analysis

The dataset shows an overall churn rate of 61.15%, indicating that most subscribers eventually terminate their memberships. The timing of churn follows a distinct pattern, with the highest risk periods being 31-90 days (27.8% of churns) and 366+ days (33.1% of churns).

4.2. Key Churn Drivers

We considered that subscription plans and payment methods are not key drivers given limited data on churn for all plans and payment methods. Yet, our analysis identified several drivers:

1. Equipment MSRP shows a strong inverse relationship with churn, where Budget equipment owners (<$500) have the highest churn rate at 83.1%, followed by Mid-range ($500-$999) at 75.8%, Premium ($1000-$1999) at 56.8%, and Luxury equipment owners (>$2000) demonstrating nearly half the churn rate at 44.1% compared to the budget segment.

2. Age shows a clear linear relationship with retention, with under-25 showing the highest churn rate (75.9%), with a decrease through each age bracket (25-34 at 70.2%, 35-44 at 61.7%, 45-54 at 55.2%).

3. Equipment type significantly influences retention, with digital-only options showing the highest churn rates (TV App 84.6%, Wearable 72.4%, Mobile App 71.7%), traditional cardio equipment in the middle range (Elliptical 67.7%, Rower 65.9%, Treadmill 60.2%, Bike 54.6%), and the premium Vault equipment demonstrating the lowest churn (47.9%).

4. Tenure: bimodal distribution for churn timing, with 27.8% between 31-90 days, and 33.1% after 366 days, suggesting critical intervention points during early onboarding and at one-year mark.

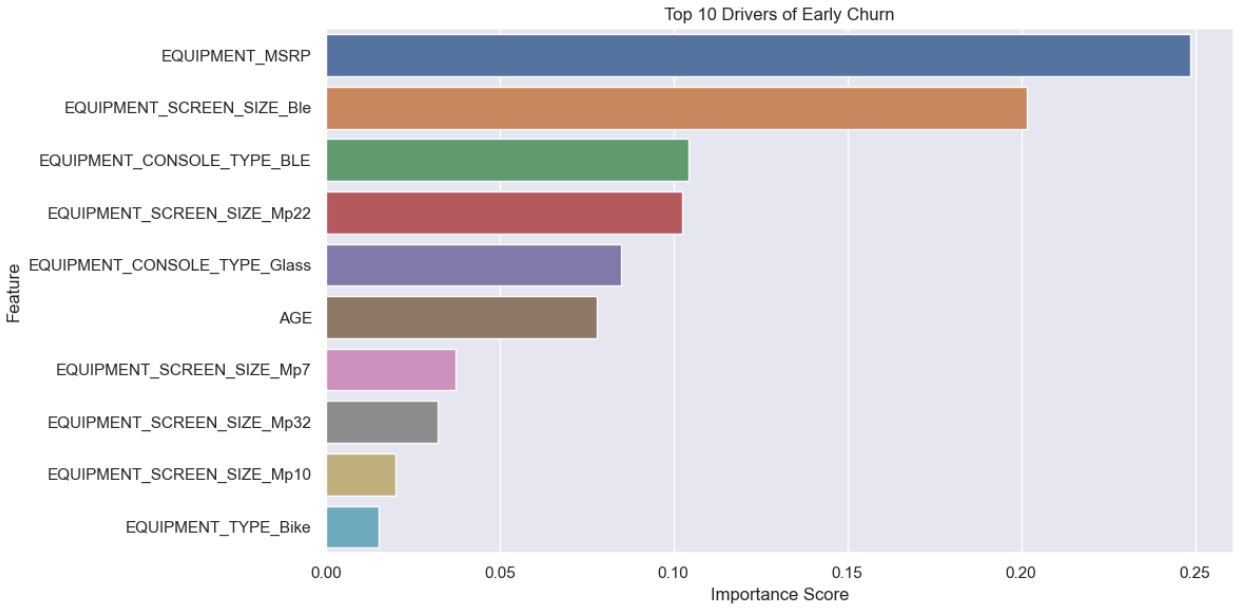

Figure 2: Top 10 Drivers of Early Churn. *This horizontal bar chart displays the most significant predictors of early subscriber churn, ranked by their importance scores. Equipment price (MSRP) emerges as the dominant factor with an importance score of approximately 0.25, suggesting that initial investment level is the strongest predictor of early departure. Screen size (Ble) follows as the second most influential factor at around 0.20. Console type (BLE and Glass) and various screen size specifications (Mp22, Mp7, Mp32, Mp10) also appear as significant contributors. Age ranks as the sixth most important factor with a score of about 0.09, indicating demographic factors play a notable but secondary role compared to equipment characteristics. Equipment type (Bike) rounds out the list with the lowest importance score among the top 10 factors. These findings suggest that early-stage retention strategies should prioritize addressing equipment investment barriers and optimizing the user experience across different console and screen configurations.*

4.3. Predictive Model Performance

Our Random Forest model demonstrated strong at-risk identification with 86.7% accuracy, 59.9% precision, 96.9% recall, 74.1% F1 score, and 0.933 AUC-ROC, showing excellent capability to identify churners.

4.4. User Personas

Through **clustering and predictive modeling,** we **identified six personas with varying engagement and churn risk:**

Persona	Segment Name	Primary Issue and Retention Challenges	Definition	Demographics	Workout Behavior	Subscription & Equipment	Churn Risk (Days)	Churn Rate (%)	Predictive Variables
Persona_3	Uncommitted Experimenter	Lacks motivation. Engagement drops within 2 months.	Tries iFit but no routine.	18-47 years old (Avg: 35.9)	Infrequent workouts, short bursts of engagement.	TRAIN plan, primarily treadmill, mid-range MSRP ($1,227).	233.8	80.3%	- Low workout frequency - High inactivity gaps - Does not engage with advanced features
Persona_5	Value-Seeker	Cost-sensitive.	High-end equipment but price-conscious	18-84 years old (Avg: 42.1)	Moderate consistency	TRAIN plan, high-end treadmill ($3,229).	342.3	48.5%	- Uses iFit features selectively - Higher equipment investment but lower session frequency - Likely to churn if price increases
Persona_4	Quick Dropout	No strong fitness habit.	Disengages rapidly.	18-77 years old (Avg: 44.8)	Minimal usage	TRAIN plan, $1,629 MSRP.	222.1	79.9%	- Short trial period before inactivity - No recurring engagement pattern - Most likely to cancel within first 2 months
Persona_1	Fitness Diehard	Highly engaged. Low churn risk.	Lives and breathes fitness, consistently uses iFit.	18-91 years old (Avg: 48.5)	Very high engagement.	PRO plan, mid-tier treadmill ($1,933 MSRP).	690.6	1.5%	- Daily workouts - High completion rates for structured programs
Persona_6	Fit Parent	Struggles to fit workouts into routine.	Health-conscious but struggles with external responsibilities.	37-100 years old (Avg: 54.0)	Moderate engagement.	TRAIN plan, bike ($1,646 MSRP).	326.3	67.3%	- Workouts on weekends/early mornings - Gaps in activity due to external responsibilities - Likely to engage with shorter workouts and pause account
Persona_2	Seasonal User	Engages heavily in Jan-Feb, churns in March.	Engages around fitness trends (e.g., New Year's resolutions).	43-100 years old (Avg: 56.5)	Highly active for short periods	TRAIN plan, treadmill ($1,499 MSRP).	265.2	69.5%	- Prefers "beginner" or challenge-based programs - Less interaction with long-term plans

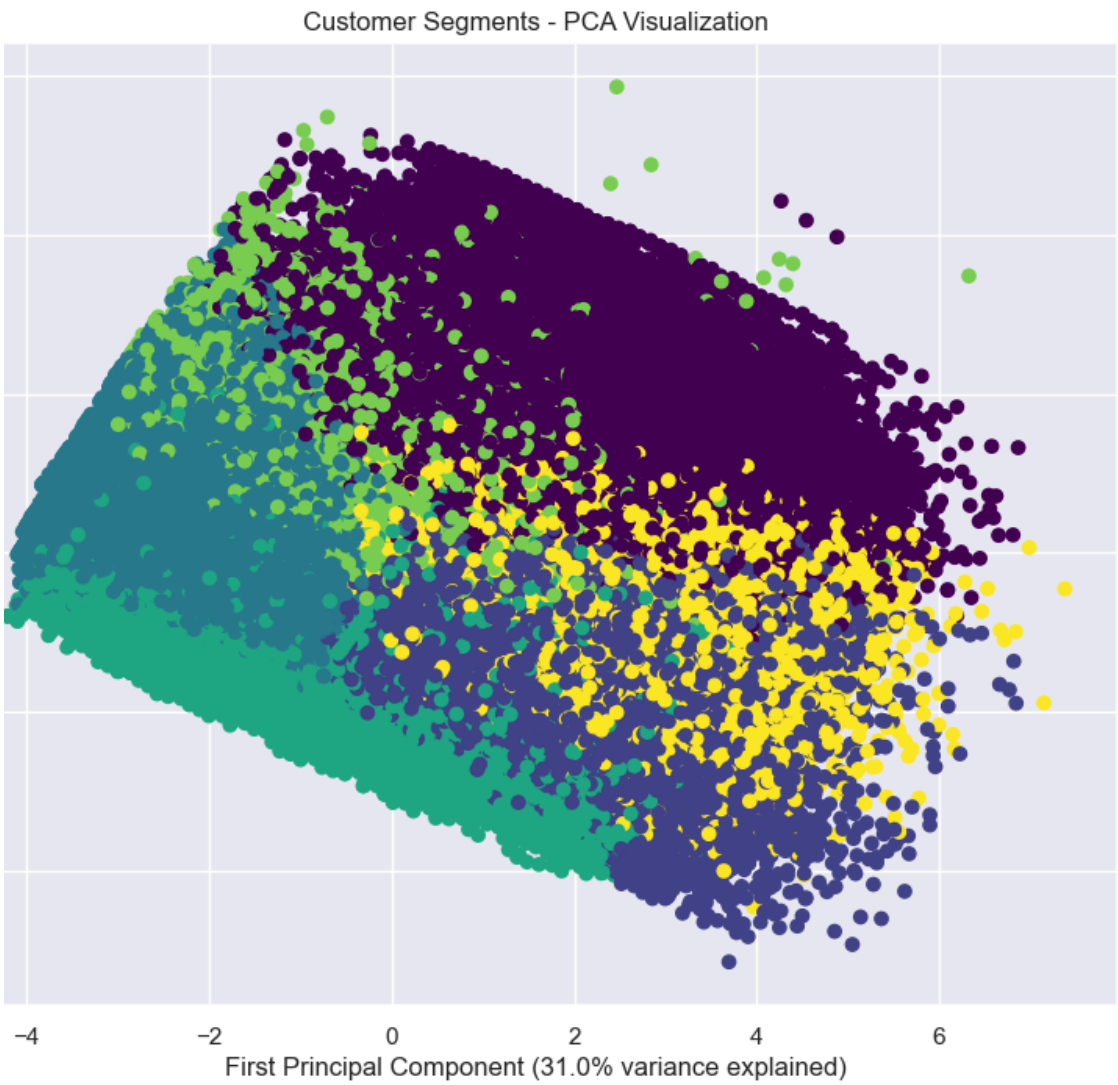

Figure 3: PCA Visualization of Customer Segments with Identified Personas. *This visualization displays customer segmentation through Principal Component Analysis, with the first two principal components explaining 52.4% of total variance (31.0% and 21.4% respectively). Each point represents a subscriber, colored by their assigned cluster (0-5). The first principal component (x-axis) primarily separates customers by age and subscription tier, with older subscribers on the right. The second principal component (y-axis) distinguishes customers based on predicted churn time and equipment investment, with higher values indicating longer retention and more expensive equipment. The distinct cluster separation reveals six customer personas: Cluster 0 (purple, upper right) represents "Fitness Diehards" with premium equipment and low churn risk; Clusters 1-2 (blue/teal, lower) show "Seasonal Users" and "Fit Parents"; Cluster 3 (green, upper left) indicates "Quick Dropouts"; and Clusters 4-5 (yellow, center-right) represent "Value-Seekers" and "Uncommitted Experimenters" with varying retention characteristics.*

4.5. Revenue Impact Analysis

We identified 11,287 high-risk subscribers (1.8% of churners) who could be retained through targeted interventions, representing $1,625,328 in annual revenue-at-risk. Depending on effectiveness, interventions could save $406,332 (25% success rate), $812,664 (50%), or $1,218,996 (75%) annually.

4.6. Survival Analysis Results

Kaplan-Meier survival analysis revealed key retention patterns requiring targeted strategies:

1. **Critical churn windows**: highest risk in first 90 days (36.7% of churns, 27.8% between 31-90 days) and at 1-year mark (33.1%).
2. **Equipment**: digital users (Mobile/TV apps) churn more than physical equipment users.
3. **Age**: clear gradient with younger subscribers churning more.

4.7. Social Media Sentiment Insights

We identified five complaint categories driving cancellation discussions:

1. **Difficulty Canceling**: users reported convoluted cancellation processes requiring phone calls.
2. **Post-Cancellation Charges**: complaints about continued billing.
3. Frustration with response times and resolution quality, particularly in billing.
4. **Membership Confusion**: unclear conditions regarding commitment periods and renewals.
5. **Legal Disputes**: discussions about potential class action lawsuits on cancellation difficulties.

Posts on cancellation demonstrated higher engagement (average 8.3 comments versus 3.1 for non-cancellation posts). While 22 posts is a limited sample, it adds important context to quantitative findings, particularly around customer friction points.

4.8. Seasonal Trends and External Factors

We identified strong seasonal patterns affecting subscriptions:

1. **New Year's Resolution Effect**: January shows 25% above-baseline search volume for "iFit," "home workouts," and "fitness app," with increased new subscriptions across 5 years.
2. **Post-Resolution Drop-Off**: March-April shows 40% decreased engagement from January and higher churn.
3. **Search-Subscription Correlation**: strong correlation ($r = 0.78$) between search volume and sign-ups confirms external interest predicts subscription behavior.
4. **Seasonal Personas**: "Seasonal Users" shows high January-February engagement and March disengagement.

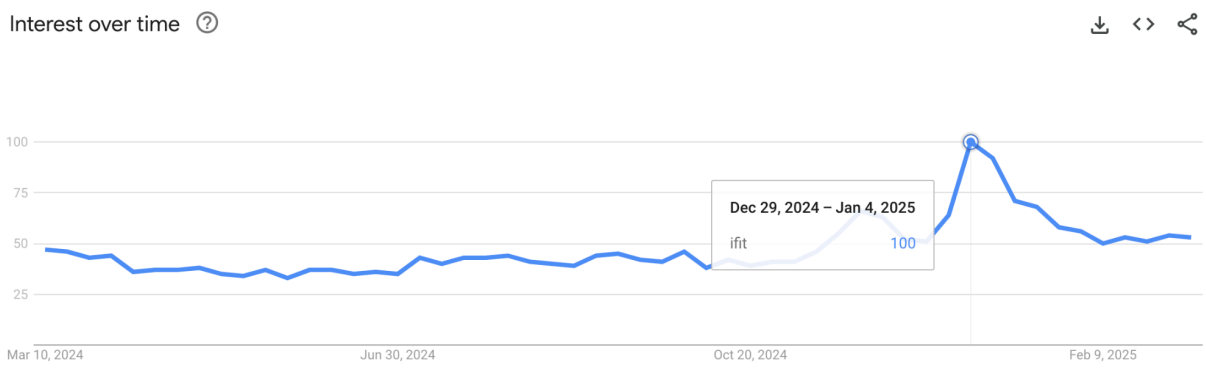

Figure 4: *Google Trends visualization for iFit searches (1 year horizon), peaking the week of New Years.*

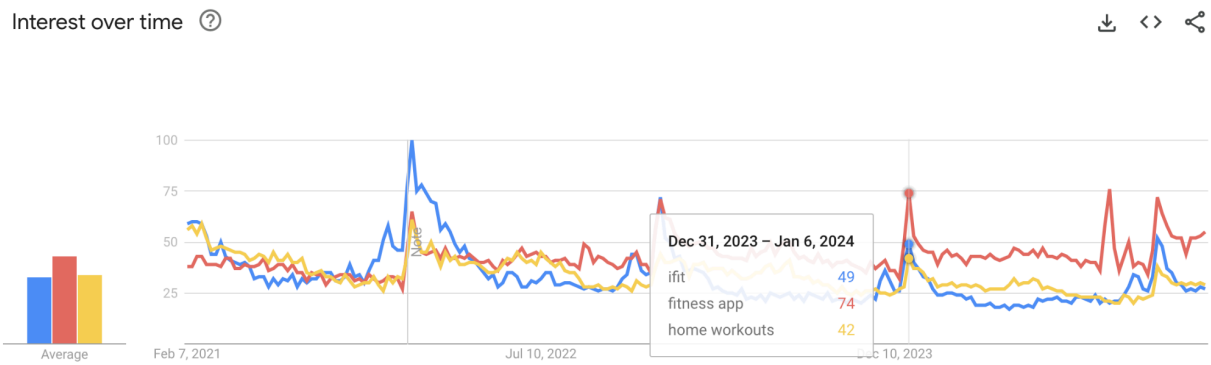

Figure 5: *Google trends for "iFit", "fitness app", and "home workouts" searches (5 year horizon), peaking yearly the week of New Years*

Sentiment analysis revealed a mixed landscape:

- **Positive** (38%): praise for workout variety, interactive coaching, and integration with equipment.
- **Neutral** (22%): General discussions, user questions.
- **Negative** (40%): Complaints about pricing, app bugs, billing issues, and poor customer support.

Negative feedback centered on **pricing concerns** (TRAIN plan perceived as overpriced for content offered), **technical frustrations** (app crashes, syncing failures), and **customer service problems** (slow response times, unresolved billing issues), providing clear targets for reducing churn.

5. CONCLUSIONS

Our iFit analysis reveals several actionable insights to improve retention and maximize LTV:

1. **Equipment investment**: luxury equipment owners churn at half the rate of budget equipment owners.
2. **Age** impacts churn. Under-35 subscribers require targeted retention strategies.

3. Our model identified 11,287 at-risk subscribers worth **$1.6 million in annual revenue.**
4. **Customer personas** have distinct behaviors: "Fitness Diehards" demonstrate exceptional retention.
5. **Technical and customer service issues** are major pain points and increase churn.
6. **Strong seasonal patterns**, with 25% higher subscriptions in January followed by 40% engagement drops in March-April, requiring targeted intervention during these periods.

We recommend enhancing **equipment strategies** due to the strong link between investment and retention. Implement trial programs for budget subscribers to try premium equipment, create upgrade pathways with discounts, and target retention efforts at disengaged budget equipment owners. For **demographic targeting**, focus on age-specific content for younger users with higher churn rates, develop social features for under-35 subscribers, and offer fitness programs tailored to different life stages.

Personalized reminder systems, AI-driven challenges, and milestone tracking for high-risk personas can reduce churn, along with price optimization and loyalty programs utilizing tiered pricing, long-term discounts, and limited-time incentives. We also suggest implementing retention campaigns in March-April to prevent seasonal drop-offs after New Year's resolutions, Developing content specifically for busy subscribers, especially parents, and featuring family-oriented workouts with shorter sessions for demanding schedules.

Based on our **persona segmentation** and **churn analysis**, we identified **targeted interventions** addressing the segment-specific challenges:

Strategy	Personas	Recommendations	Expected Impact
Technical Fixes	All Users	Improve **device syncing, crash reduction, and workout tracking accuracy**.	Decrease **frustration-driven churn** by 10-15%.
Customer Service Enhancements	All Users	Faster response times, improved resolution of billing issues, and AI support.	Increase **customer satisfaction**, reduce complaints-related churn by 8-12%.
Offers for Budget-Conscious Users	**"Value-Seeker"**	Introduce **tiered pricing, loyalty rewards, and discounts** for long-term plans.	Reduce churn by 15% in this price-sensitive group.
Engagement Campaigns for New Year's Joiners	**"Seasonal User"**	- Launch **6-Month Transformation Plan** to sustain engagement beyond Q1. - Introduce **gamified retention challenges**.	Extend engagement **past March-April** by 10-20%.
Long-Term Commitments	**"Fit Parent," "Value-Seeker"**	- Higher annual discounts. - Loyalty bonuses after **12 months.**	Boost **long-term subscription retention** by 15-20%.
AI-Powered Workouts	**"Uncommitted Experimenter," "Quick Dropout"**	Suggest workouts tailored to engagement patterns.	Increase **completion rates** and reduce churn.
Flexible Workouts	**"Fit Parent"**	- Shorter, high-impact **15-20 min workouts**. - **"Family-friendly fitness plans"** for shared engagement.	Improve **retention in busy users**.
Seasonal Retention Strategies	**"Seasonal User"**	**March-April engagement campaigns** to **counteract post-New Year's drop-offs**.	Reduce **seasonal churn by 10%**.
Expanding Data Analysis & CLV	All Users	- Include **other social platforms (YouTube, Twitter, Trustpilot)**.	Increase **prediction accuracy and intervention success**.

Finally, we recommend implementing ML models to **proactively identify at-risk subscribers,** with intervention strategies based on risk and persona. To measure effectiveness, iFit should implement A/B testing, ongoing monitoring of churn metrics by segment, and calculate ROI of retention programs.

Additionally, several additional explorations could improve our understanding and prediction of churn:

1. **Engagement analysis**: incorporate workout frequency, content preferences, and app usage patterns to identify pre-churn warning signs.
2. **Competitive market factors**: Assess how competitor offerings affect churn decisions.
3. **Customer journey**: Analyze subscriber lifecycle to identify critical churn-risk moments.
4. **Enhanced financial modeling**: Develop sophisticated LTV models including variable costs and cross-selling.

Implementing these recommendations can reduce churn in high-risk segments, preserving revenue and customer LTV. By tackling key drivers, iFit can shift from reactive to proactive retention, creating a sustainable competitive advantage.

We grant permission for this project to be kept and used as an example for future classes.

Appendix

Table 1: Churn Distribution by Subscription Tier and Payment Method

This table shows a striking pattern in churn distribution across subscription tiers, where PREMIUM and PRO members show 0% churn rate, while the TRAIN tier experiences an extremely high 96.5% churn rate (641,979 out of 664,967 subscribers).

Subscription Tier	Not Churned	Churned	Total	Churn Rate
PREMIUM	93,523	0	93,523	0.0%
PRO	291,320	0	291,320	0.0%
TRAIN	22,988	641,979	664,967	96.5%

Table 2: Churn Distribution by Subscription Payment

This table shows a striking pattern in churn distribution across payment methods, where MONTHLY subscribers show 0% churn, ANNUAL subscribers have a modest 5.6% churn rate, while MULTI_YEAR subscribers demonstrate a very high 94.4% churn rate.

Payment Method	Not Churned	Churned	Total	Churn Rate
MONTHLY	195,901	0	195,901	0.0%
ANNUAL	174,321	10,292	184,613	5.6%
MULTI_YEAR	37,609	631,687	669,296	94.4%

Table 3: Churn Rate by Equipment Tier.

Equipment Tier	Churn Rate
Budget (<$500)	83.1%
Mid-range ($500-$999)	75.8%
Premium ($1000-$1999)	56.8%
Luxury (>$2000)	44.1%

Table 4: Churn Rate by Age Group.

Age Group	Churn Rate
<25	75.9%
25-34	70.2%
35-44	61.7%
45-54	55.2%
55+	51.8%

Table 5: Churn Rate by Equipment Type.

Equipment	Churn Rate
TV App	84.6%
Wearable	72.4%
Mobile App	71.7%
Elliptical	67.7%
Rower	65.9%
Treadmill	60.2%
Bike	54.6%
Vault	47.9%

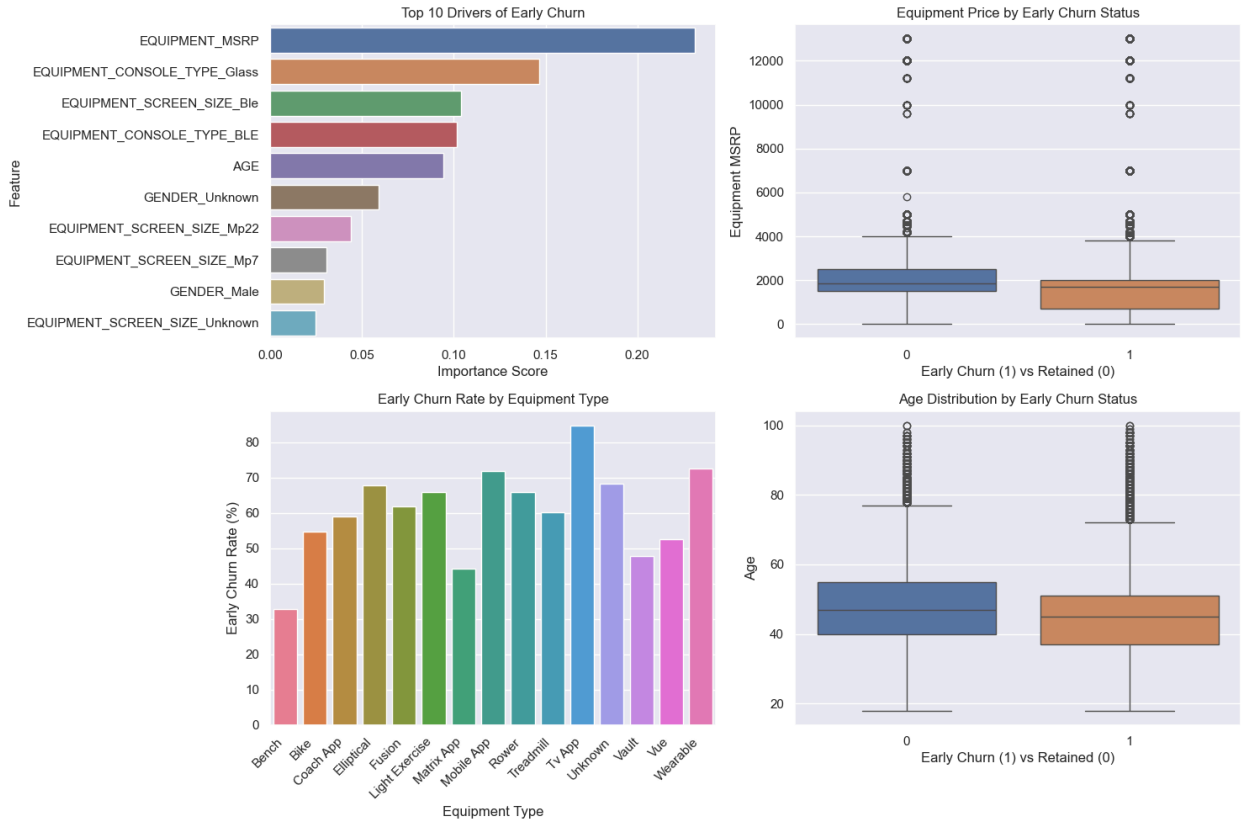

Figure 1: Key Churn Factors and Demographics Analysis. *This four-panel visualization reveals key insights about churn factors: (Top left) Equipment characteristics emerge as the strongest predictors of churn, with equipment price (MSRP) being the most influential factor, followed by console type and screen size; (Top right) Box plot showing equipment price distribution by churn status, with retained customers (0) typically owning higher-priced equipment than churned customers (1); (Bottom left) Churn rates by equipment type, with TV Apps showing the highest churn rate (~84%) while Bench equipment shows the lowest (~32%), demonstrating the correlation between equipment investment and retention; (Bottom right) Age distribution by churn status, highlighting some outliers but suggesting age plays a role in retention probability.*

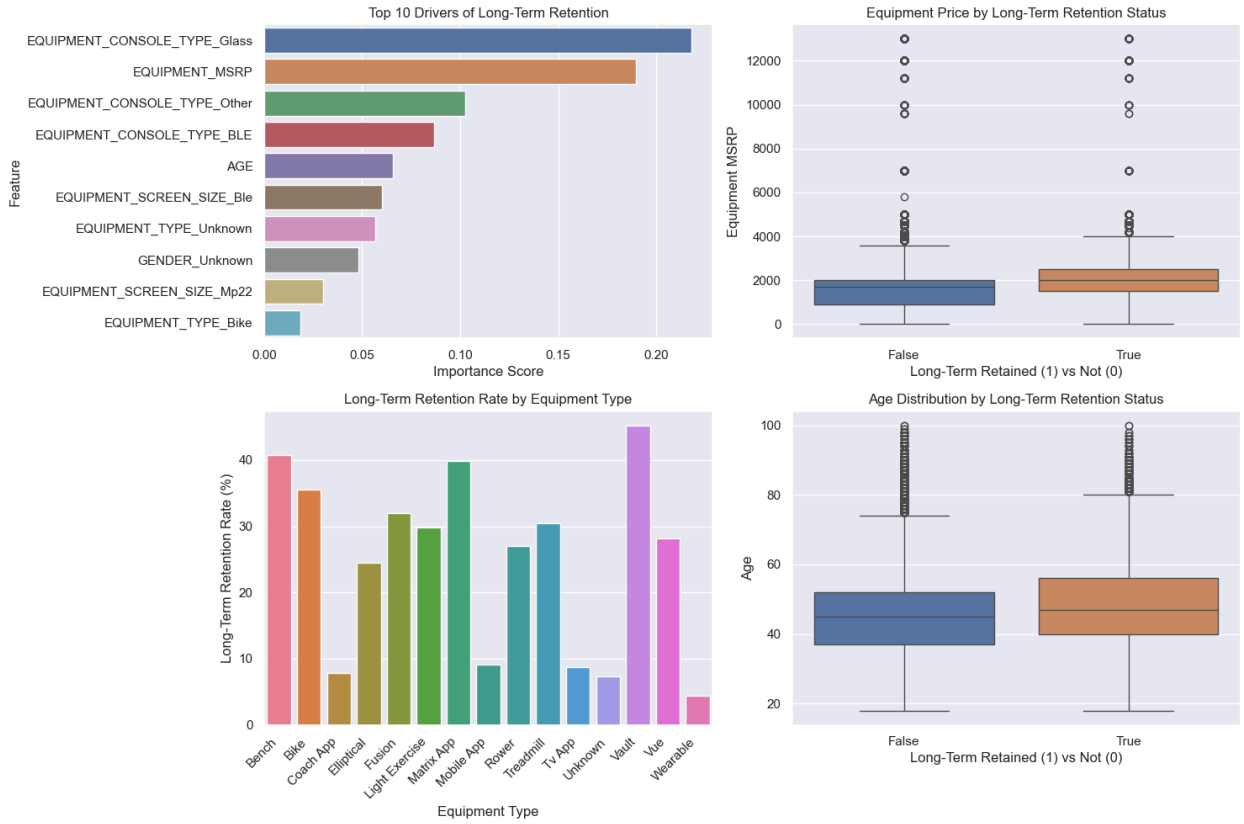

Figure 2: Retention Drivers Analysis. *This four-panel visualization examines the drivers of long-term subscriber retention: (Top left) Console type, particularly Glass consoles, and equipment price (MSRP) emerge as the strongest predictors of long-term retention, followed by other equipment characteristics and age; (Top right) Box plot comparing equipment price between long-term retained customers (True) and shorter-term customers (False), showing long-term users typically own higher-priced equipment with a median price around $2,000 versus $1,500 for non-retained users; (Bottom left) Long-term retention rates by equipment type, showing Vault equipment with the highest retention (~45%) followed by Bench (~40%) and Light Exercise equipment (~40%), while Wearable devices show the lowest long-term retention (~4%); (Bottom right) Age distribution by retention status, indicating long-term retained customers (True) tend to be older than short-term users (False), with higher median and upper quartile ages. This analysis confirms that premium equipment investment and older age demographics are significant predictors of long-term customer loyalty.*

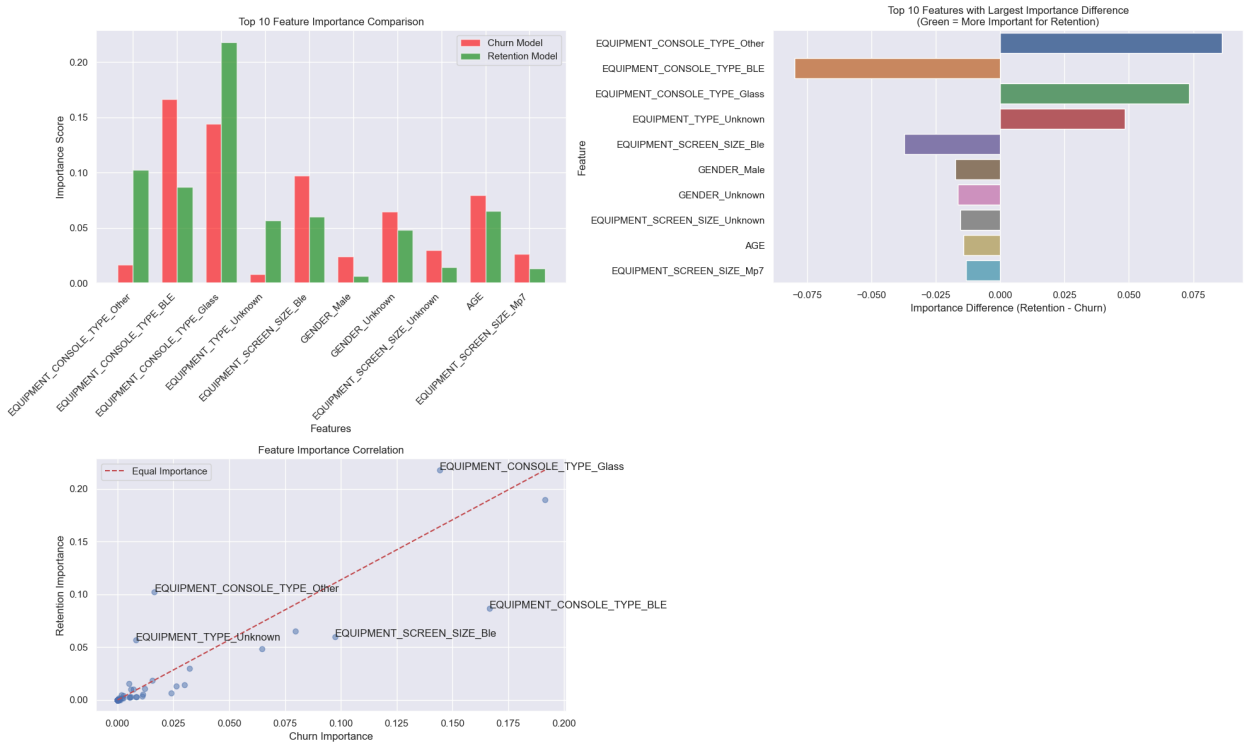

Figure 3: Comparative Analysis of Feature Importance in Churn and Retention Models. *This three-panel visualization reveals distinct feature importance patterns across churn and retention models: (Top left) Bar comparison showing significant differences in how features rank in importance between models, with Glass console type being exceptionally important for retention (0.22) but less so for churn (0.14), while BLE console type shows the opposite pattern; (Top right) Importance difference plot highlighting features with greater impact on retention (positive values) versus churn (negative values), with console types showing the most dramatic differences - Other and Glass consoles strongly associated with retention, while BLE consoles more strongly predict churn; (Bottom) Correlation plot reveals moderate consistency in feature importance between models, with some notable deviations from the diagonal line indicating features that play substantially different roles in retention versus churn prediction. This analysis suggests equipment characteristics influence customer behaviors differently depending on whether we're examining factors that promote loyalty versus those that trigger departures.*

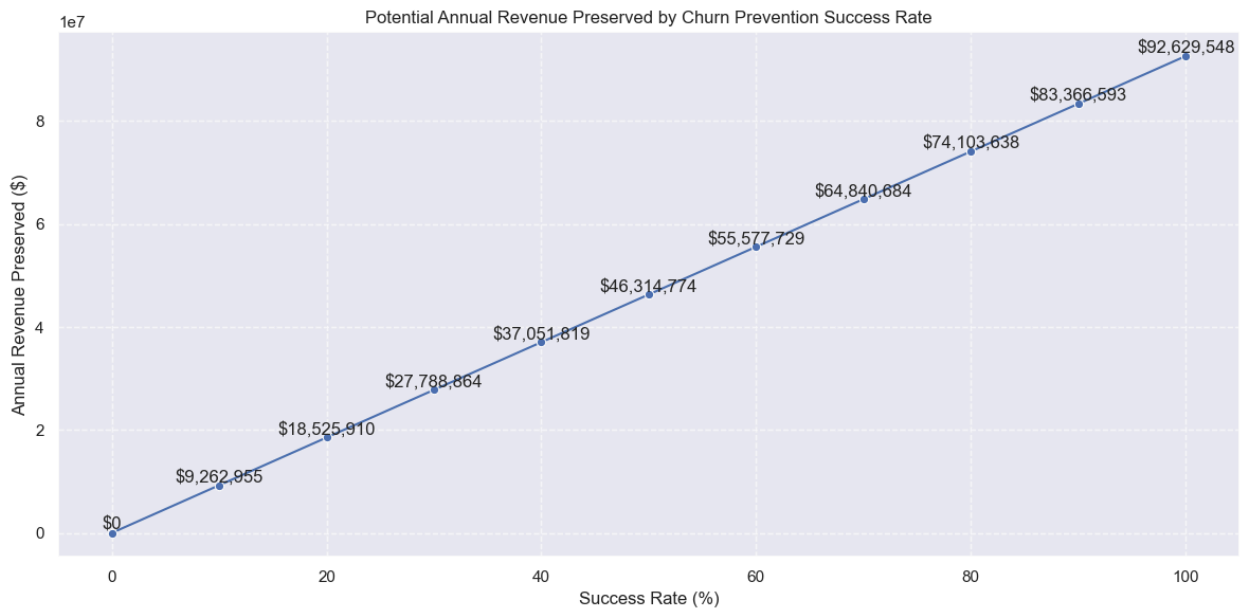

Figure 4: Potential Annual Revenue Preserved by Churn Prevention Success Rate. This line graph illustrates the projected annual revenue that could be preserved through various levels of churn prevention effectiveness. The x-axis represents the success rate of retention initiatives (0-100%), while the y-axis shows the corresponding annual revenue preserved in dollars. The relationship is linear, with each 10% improvement in success rate generating approximately $9.3 million in additional preserved revenue. At modest success rates (10-30%), the company could save $9.3-27.8 million annually, while moderate success (40-60%) could preserve $37.1-55.6 million. Higher effectiveness levels (70-90%) could save $64.8-83.4 million, with a perfect 100% success rate potentially preserving $92.6 million annually. This visualization highlights the substantial financial impact that even incremental improvements in retention strategies could deliver to iFit's bottom line.

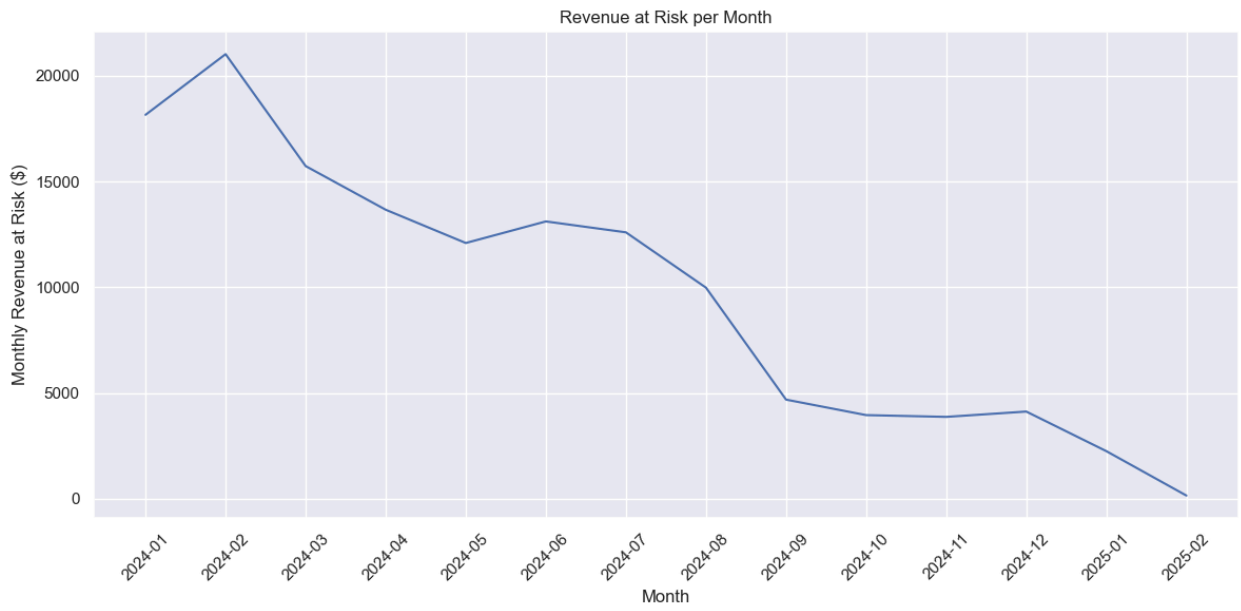

Figure 5: Monthly Revenue at Risk Due to Predicted Churn. This line graph illustrates the projected monthly revenue at risk due to predicted subscriber churn over a 14-month period (January 2024 - February 2025). The trend shows significant seasonality, with the highest revenue at risk occurring in February 2024 (approximately $20,000), followed by a general downward trajectory throughout the year. There's a notable drop between July and September 2024, when monthly revenue at risk decreases from about $10,000 to $5,000. The lowest risk appears in February 2025, approaching zero. This pattern aligns with the previously identified seasonal subscription behaviors, where many users join during New Year's resolution periods but subsequently churn, creating cyclical revenue vulnerability that requires targeted retention strategies at specific times of the year.

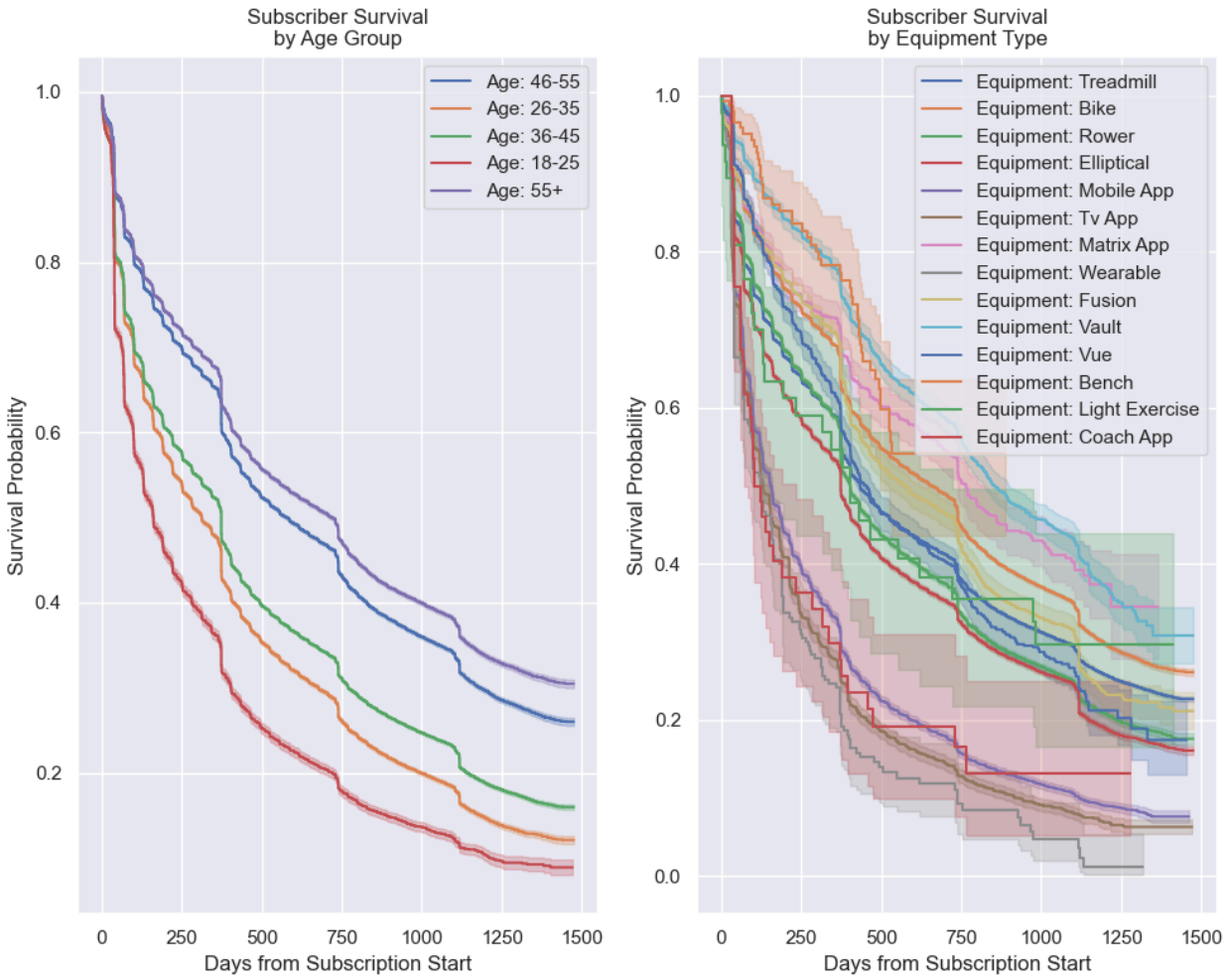

Figure 6: Kaplan-Meier Survival Curves by Age Group and Equipment Type. *This dual-panel visualization shows subscriber retention patterns over time: (Left) Survival curves by age group demonstrate a clear age gradient in retention, with older subscribers (55+, 46-55) showing significantly higher survival probabilities throughout the subscription lifecycle compared to younger cohorts (18-25, 26-35). At the 500-day mark, approximately 55% of 55+ subscribers remain active versus only 25% of 18-25 year olds. (Right) Survival curves by equipment type reveal substantial variation in retention based on equipment investment, with premium equipment types (Vault, Bike) associated with higher retention rates, while digital-only options (TV App, Mobile App) show the steepest decline. The shaded confidence intervals illustrate statistical reliability across different equipment segments. Both panels highlight critical intervention periods, with steep drops in the first 250 days followed by more gradual decline, suggesting different retention strategies may be needed at different lifecycle stages.*

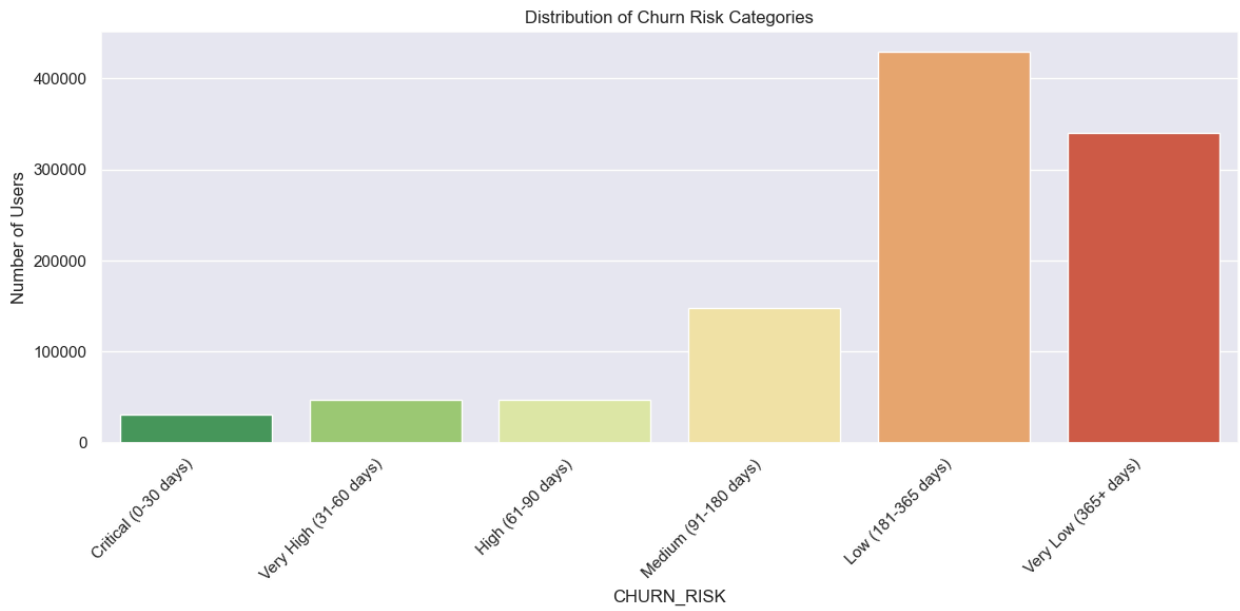

Figure 7: Distribution of Churn Risk Categories. *This bar chart illustrates the distribution of iFit subscribers across six churn risk segments based on predicted time-to-churn. The largest segment comprises "Low" risk users (181-365 days until predicted churn) with over 400,000 subscribers, followed by "Very Low" risk users (365+ days) with approximately 340,000 subscribers. Medium-risk users (91-180 days) represent about 150,000 subscribers, while high-risk categories together account for fewer subscribers: "High" risk (61-90 days) and "Very High" risk (31-60 days) each with around 45,000 subscribers, and "Critical" risk (0-30 days) with approximately 30,000 subscribers. This distribution highlights the opportunity to target preventive interventions at the smaller high-risk segments to maximize retention efficiency.*

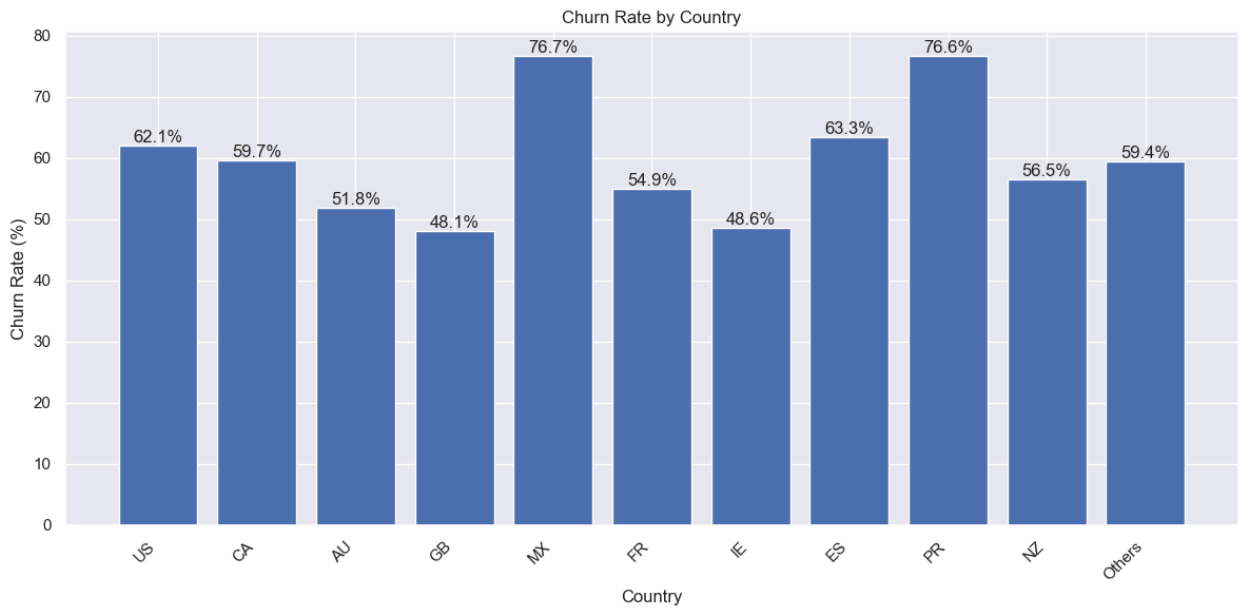

Figure 8: Churn Rate Comparison Across Major Markets

This bar chart displays the subscriber churn rates across iFit's key international markets. Mexico (MX) and Puerto Rico (PR) show the highest churn rates at approximately 76.7% and 76.6% respectively. Spain (ES) follows with 63.3%, while the United States (US) shows a 62.1% churn rate, slightly above Canada (CA) at 59.7%. Several markets demonstrate notably better retention, with Great Britain (GB) and Ireland (IE) having the lowest churn rates at 48.1% and 48.6% respectively, followed by Australia (AU) at 51.8%. France (FR), New Zealand (NZ), and other markets show moderate churn rates between 54.9% and 59.4%. These significant geographic variations suggest that market-specific factors such as cultural attitudes toward fitness, local competition, or regional content relevance may significantly influence customer retention, highlighting the need for regionally-tailored retention strategies.

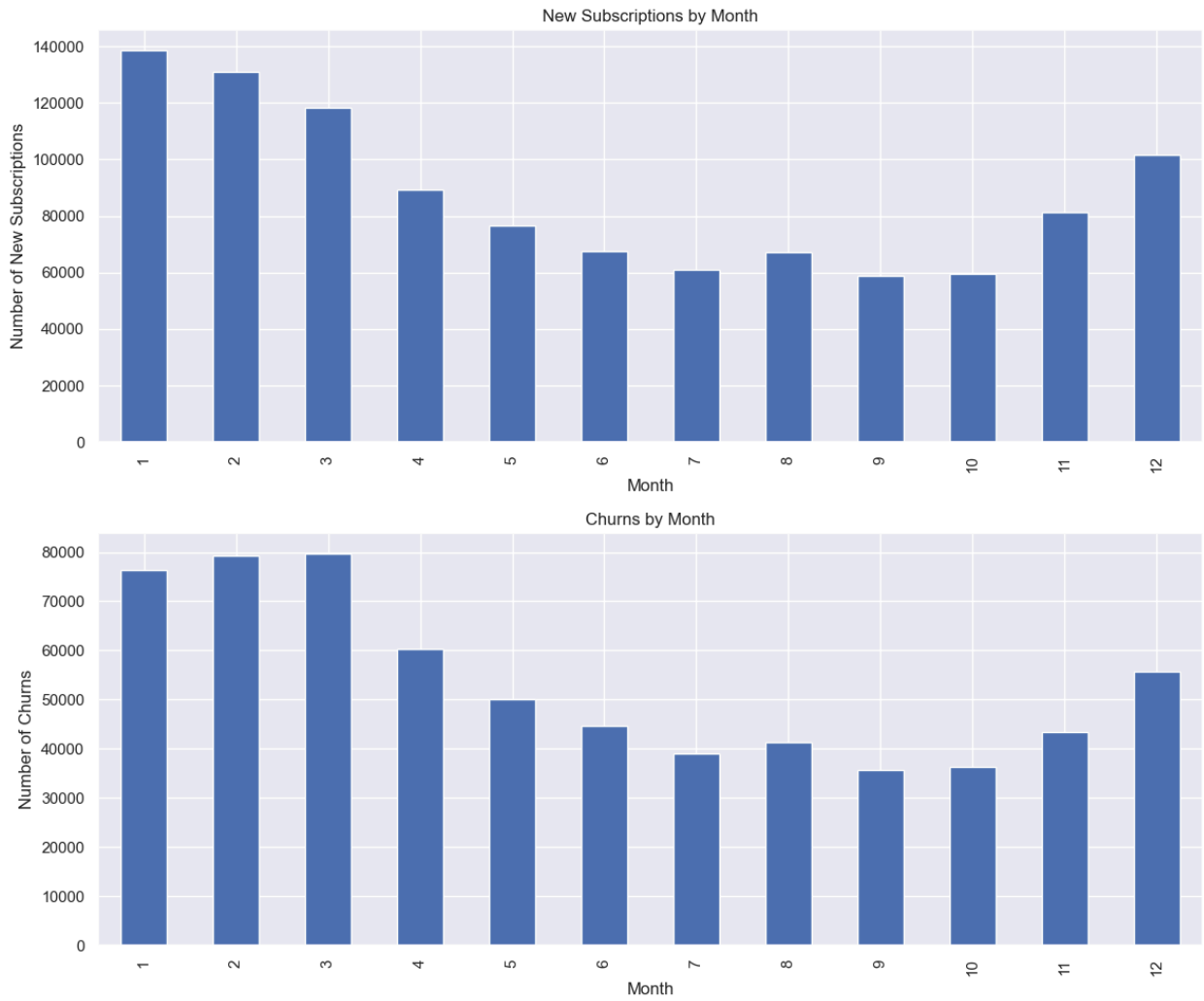

Figure 9: Seasonal Patterns in Subscription Acquisition and Churn

This dual-panel visualization reveals clear seasonal patterns in iFit's subscription lifecycle: (Top) New subscriptions peak in January (~140,000) and gradually decline through Q1, reaching their lowest points in September-October (~60,000) before climbing again in November-December, reflecting the strong influence of New Year's resolutions and holiday gift-giving; (Bottom) Churn volume shows a similar but slightly delayed pattern, with highest cancellations occurring in February-March (~80,000), presumably as New Year's resolution subscribers disengage, followed by a mid-year trough in July-October (~35,000-40,000), before rising again in Q4. This clear visualization of the "resolution effect" highlights the importance of targeted retention strategies during critical periods, particularly in Q1 when both acquisition and subsequent loss are at their highest.

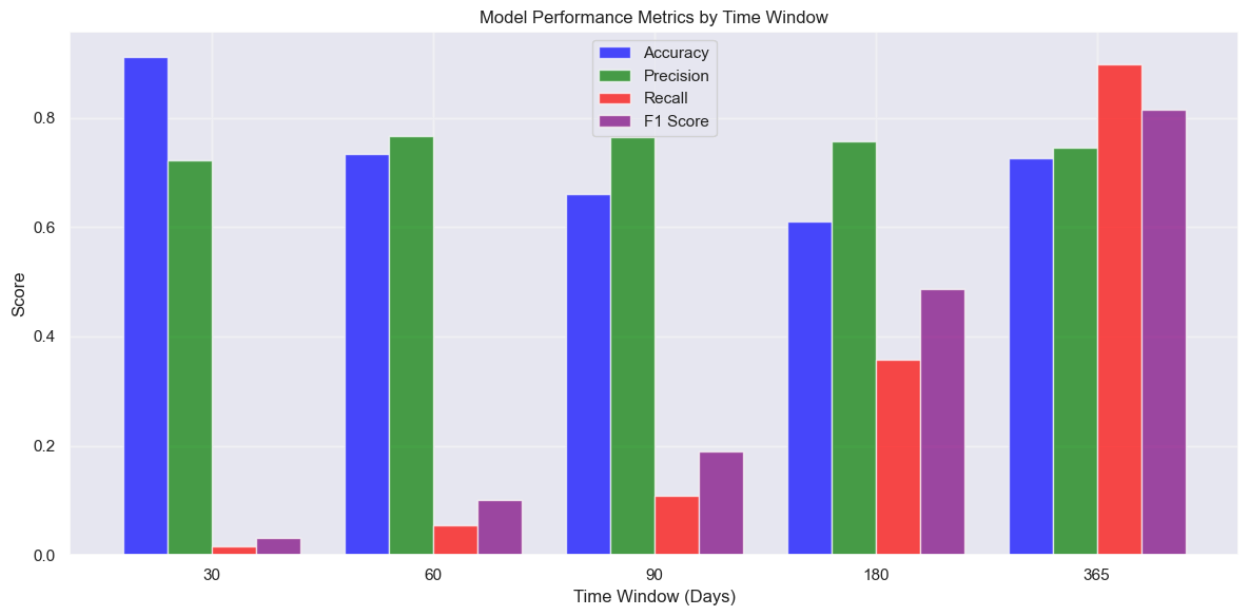

Figure 10: Predictive Model Performance Across Different Time Windows. *This bar chart compares the performance metrics of the churn prediction model across five different prediction windows (30, 60, 90, 180, and 365 days). The metrics show distinct patterns: Accuracy (blue) is highest for the 30-day window (~90%) and gradually decreases for longer prediction periods. Precision (green) remains relatively stable across all time windows (~75%), indicating consistent reliability in positive predictions. Recall (red) shows a dramatic improvement with increasing time windows, from nearly 0% at 30 days to over 90% at 365 days, suggesting the model excels at identifying long-term churn risks but struggles with very short-term predictions. F1 Score (purple), which balances precision and recall, consequently improves significantly from 30 days (~3%) to 365 days (~80%). This visualization suggests that the model is most effective for predicting churn within the 180-365 day range, where it achieves the best balance of accuracy, precision, and recall.*